Brain Tumor MRI Classification Model

Final Project – MPHY 6120 AI in Medicine

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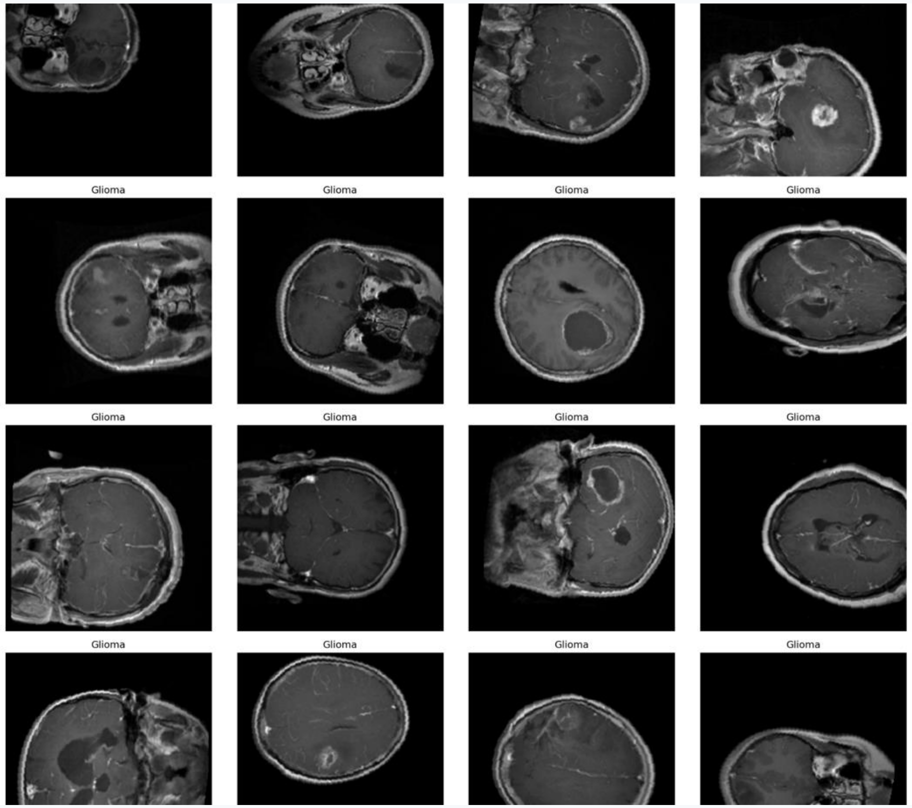
THE PROBLEM

The Clinical Need

The tool is designed to support image review, not replace clinical interpretation.

Clinical need

- Glioma, meningioma, and pituitary tumors can require different clinical management.
- MRI review depends on expert interpretation and can vary with image quality, slice selection, and reader experience.
- A consistent AI second reader may help triage cases, support education, and provide an additional check during review.



Representative MRI slices from the training split

Dataset

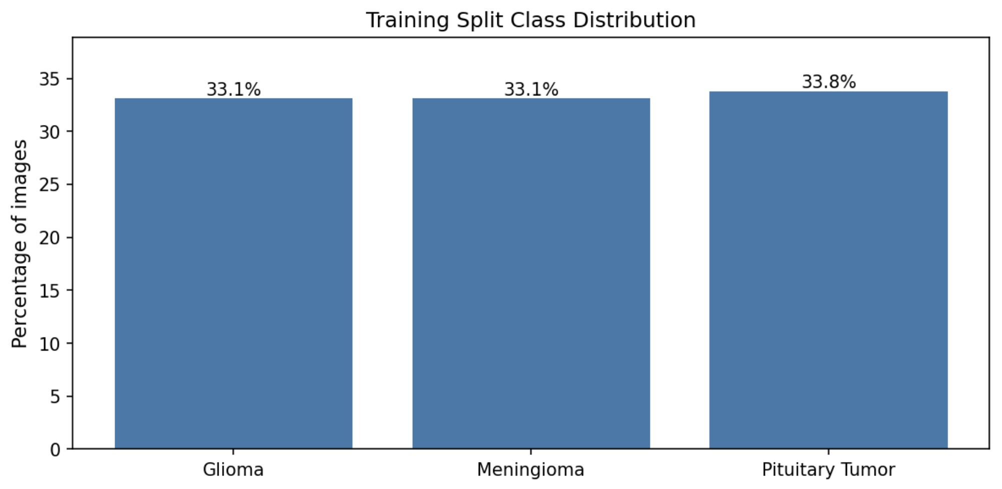
Single 2D MRI slices were used to build and evaluate a three-class classifier.

4,239
training images

908
validation images

909
test images

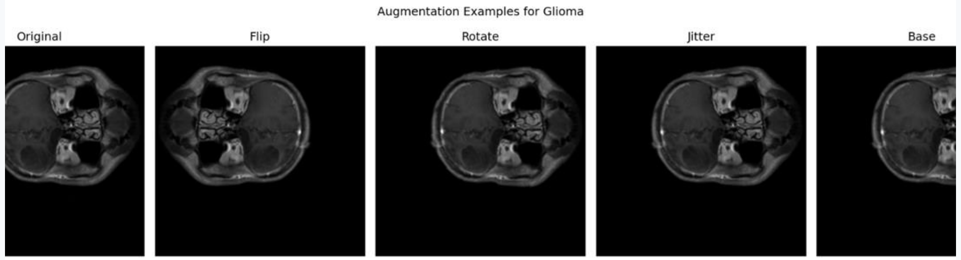
3
classes



Training split class distribution

Data decisions

- Classes: glioma, meningioma, and pituitary tumor.
- Dataset was deduplicated.
- Images were normalized and resized for model input.
- Augmentation was used to reduce overfitting and teach the model to ignore small acquisition differences.
- Evaluation was performed on a held-out test set.

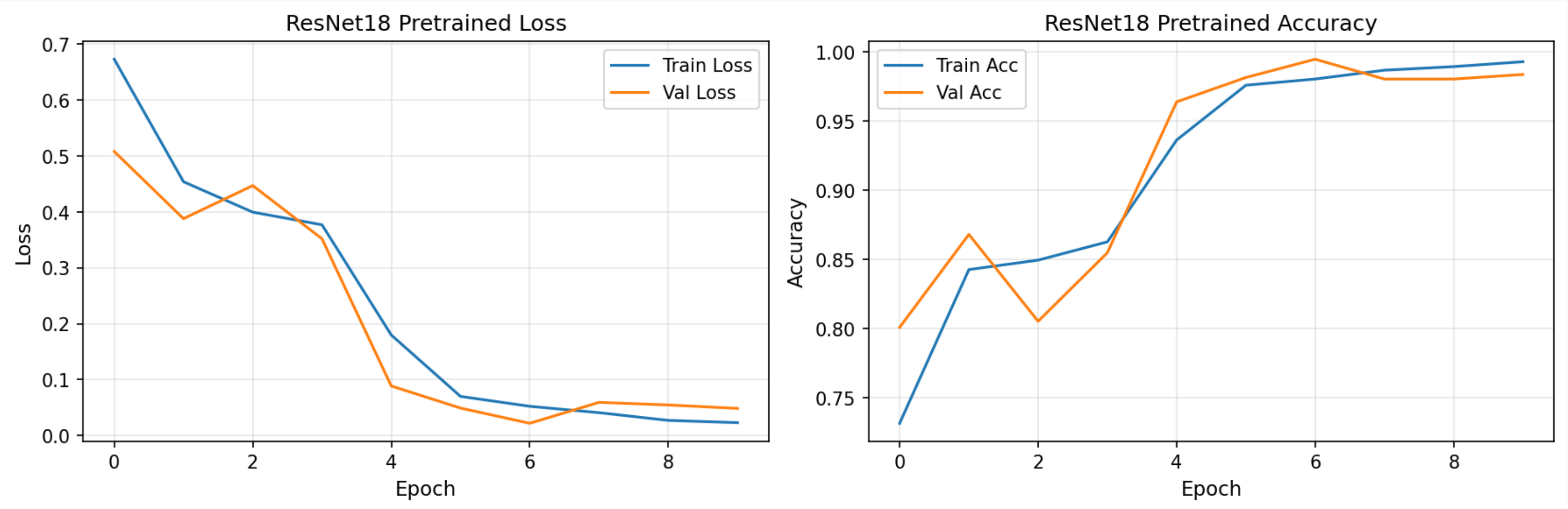


Examples of augmentation applied to training images

The Model - ResNet18 transfer learning classifier

Technical choices were kept practical: pretrained CNN, limited training time, and multiple explanation checks.

MRI slice → Preprocess + augment → ResNet18 pretrained → Class + confidence



Loss -Accuracy Curve for Train and Validation dataset for ResNEt18 Pretrained

Key technical decisions: transfer learning, class-balanced evaluation, confidence calibration review, and two interpretability methods.

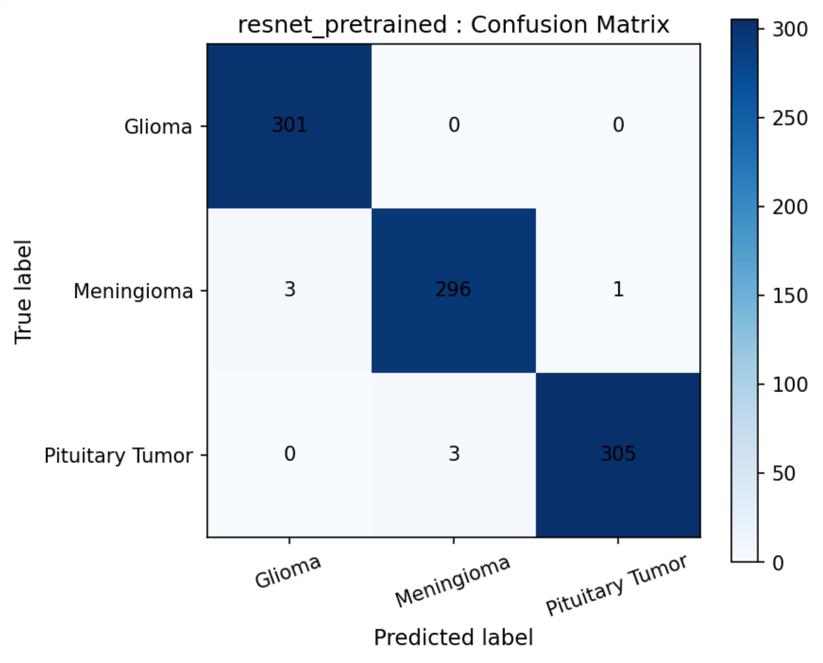
Test Performance

The model performed well on the internal test set, but errors cluster around meningioma cases.

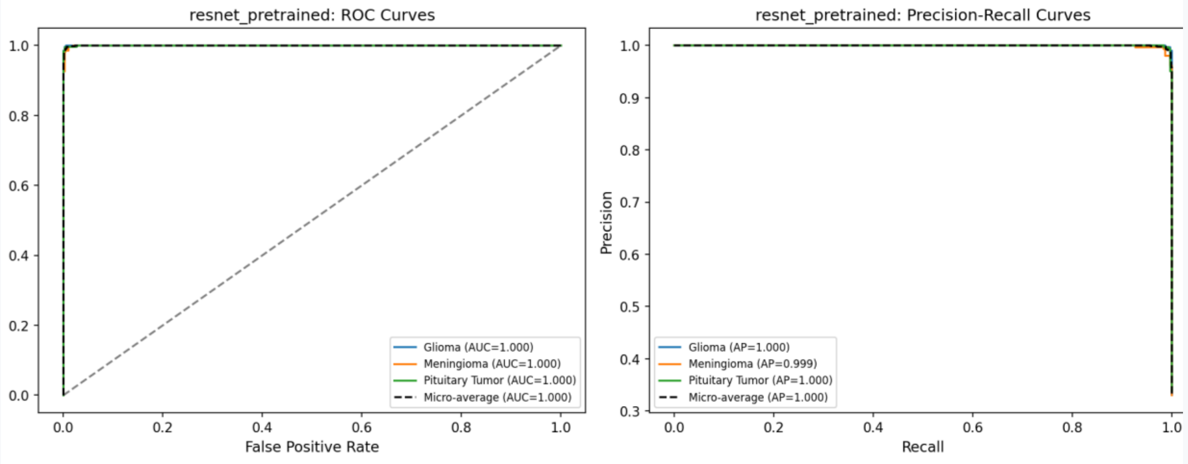
0.99
Macro Accuracy

1.000
ROC-AUC
all classes ≈ 1.00

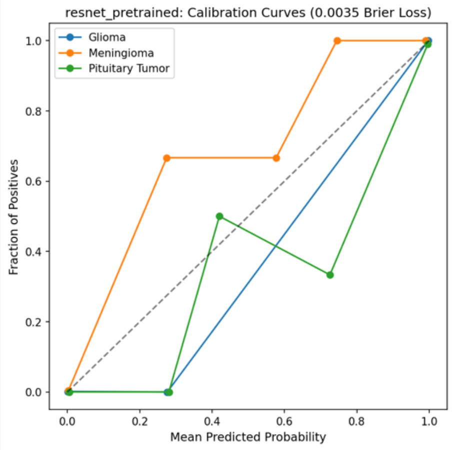
0.0035
Brier loss
lower is better



Confusion matrix on held-out test set



ROC and precision-recall curves



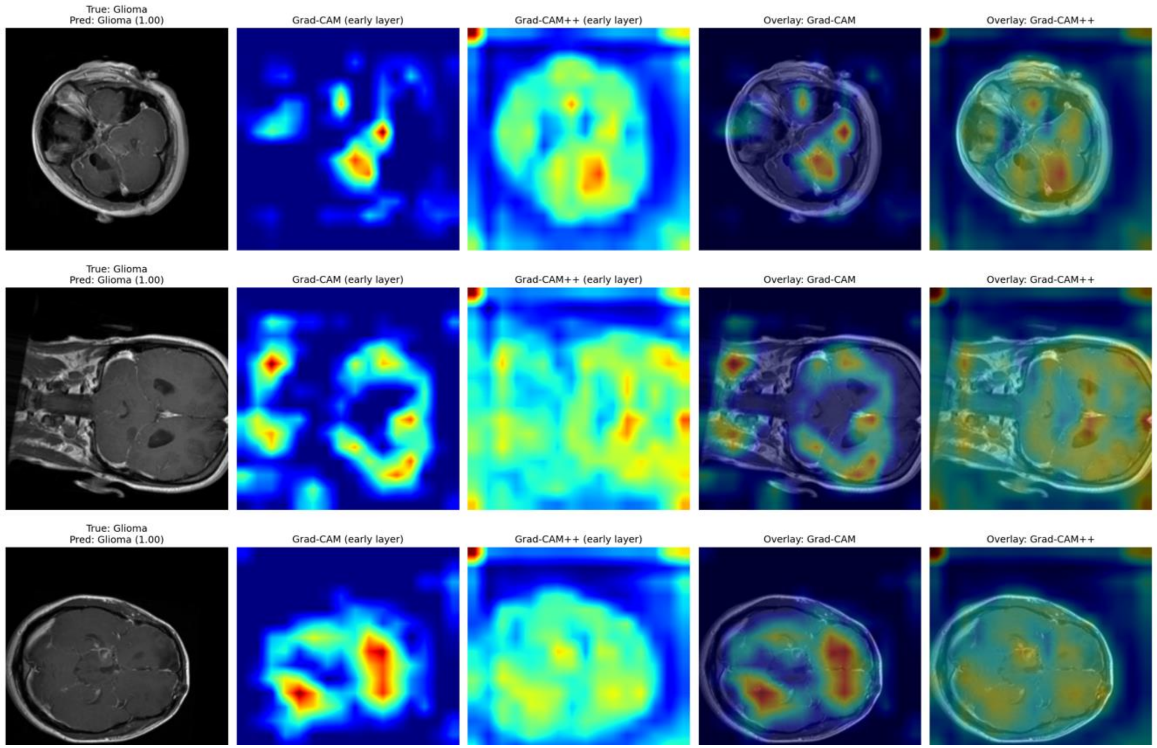
Calibration curve

Assessment

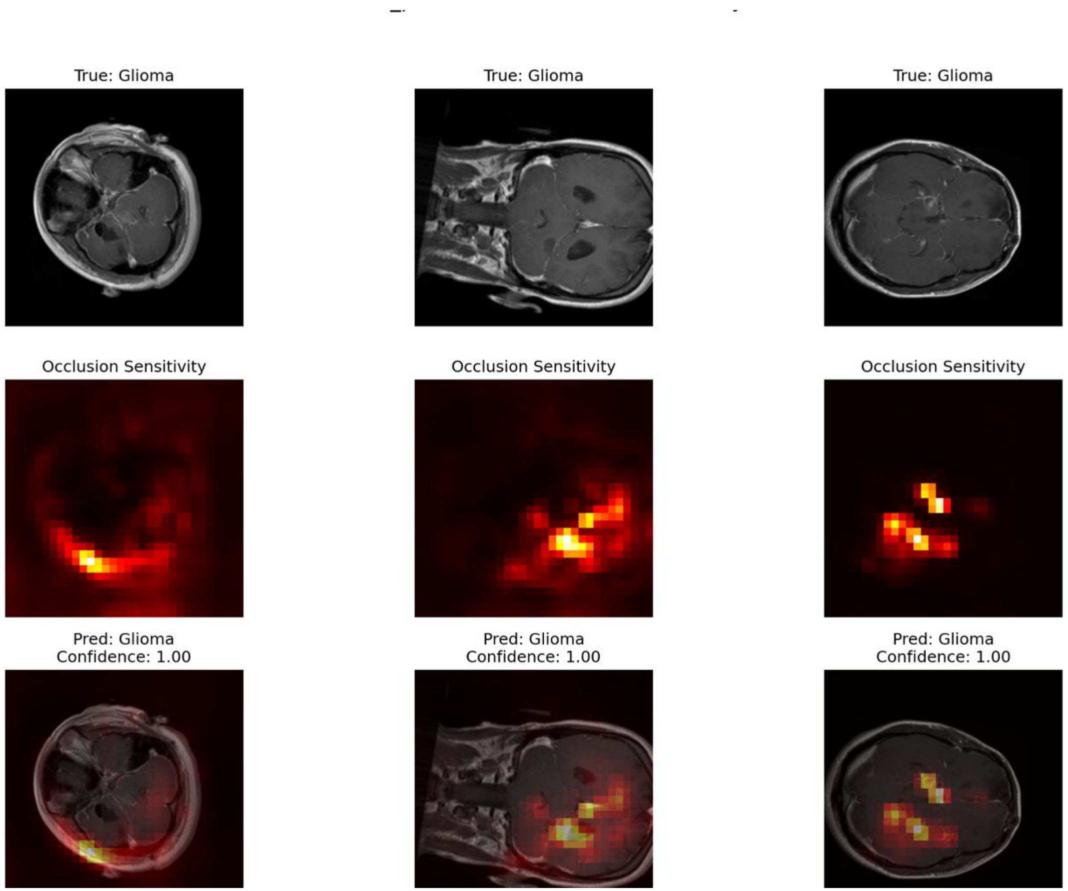
- Errors are rare but clinically meaningful.
- Confidence is often very high, so calibration must be monitored.
- External validation is still required.

Interpretability

Grad-CAM and occlusion were used to test whether the model looked at clinically reasonable regions.



Grad-CAM on test dataset

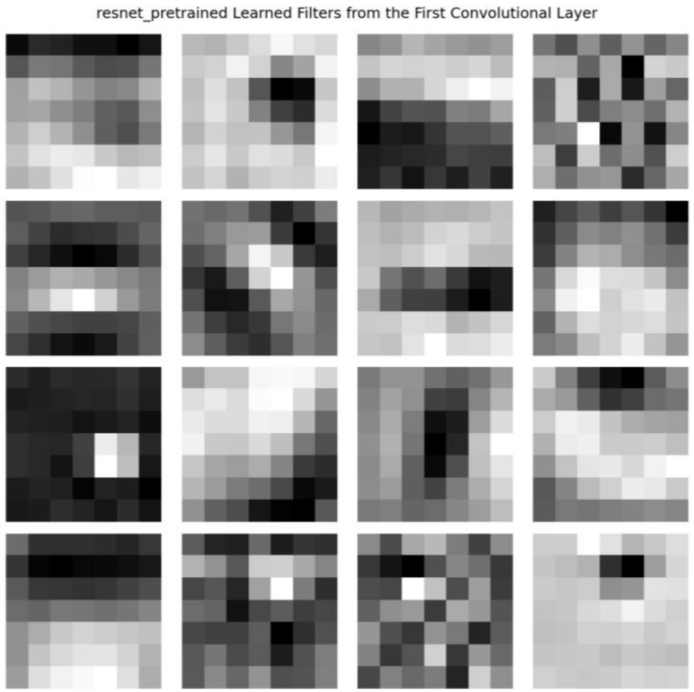


Occlusion on test dataset

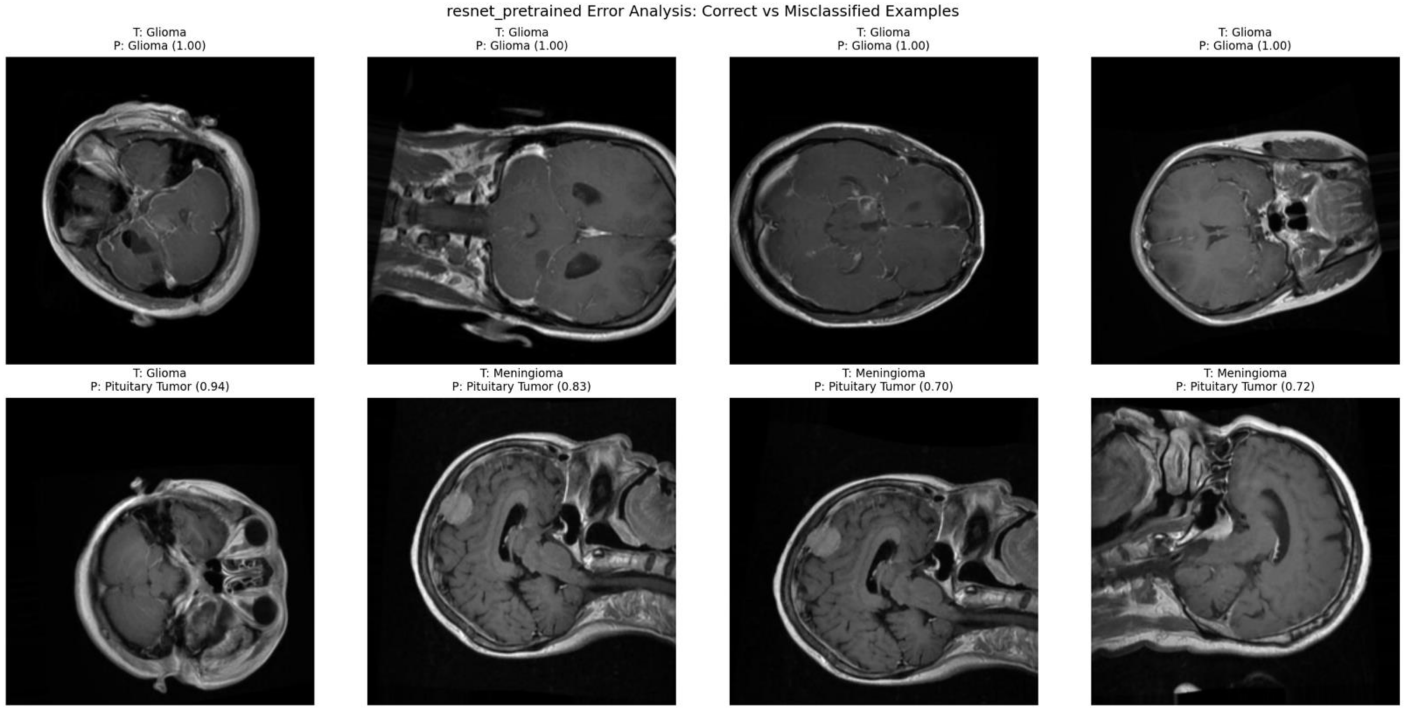
RESULTS

Interpretability

Grad-CAM and occlusion were used to test whether the model looked at clinically reasonable regions.



Filters



Correct vs misclassified examples

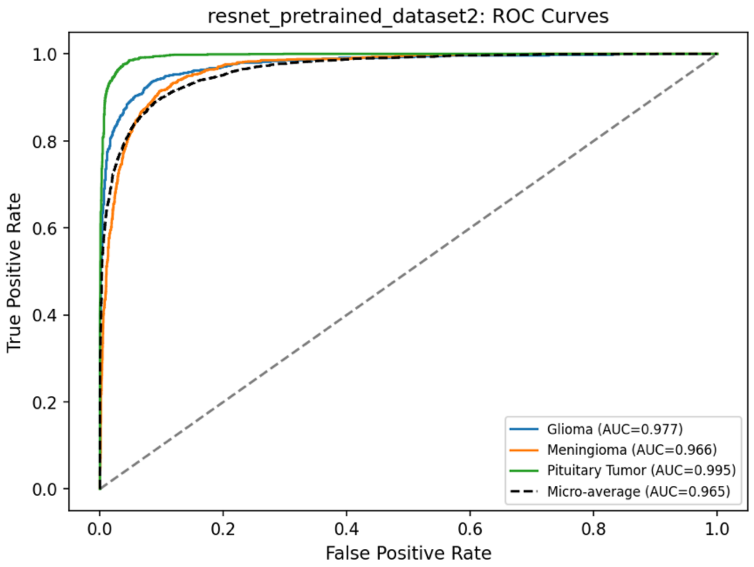
Test Performance – Additional Test Data (Glioma and Meningioma only)

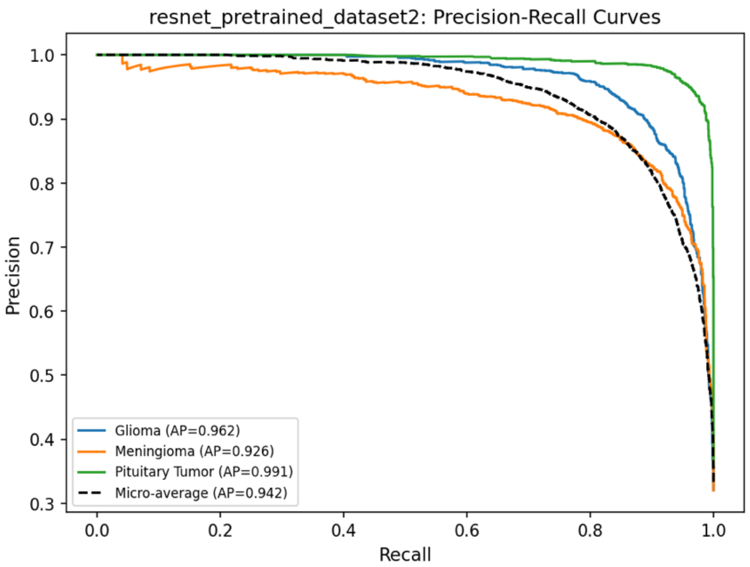
The model performed well on the internal test set, but errors cluster around meningioma cases.

0.86
Macro Accuracy

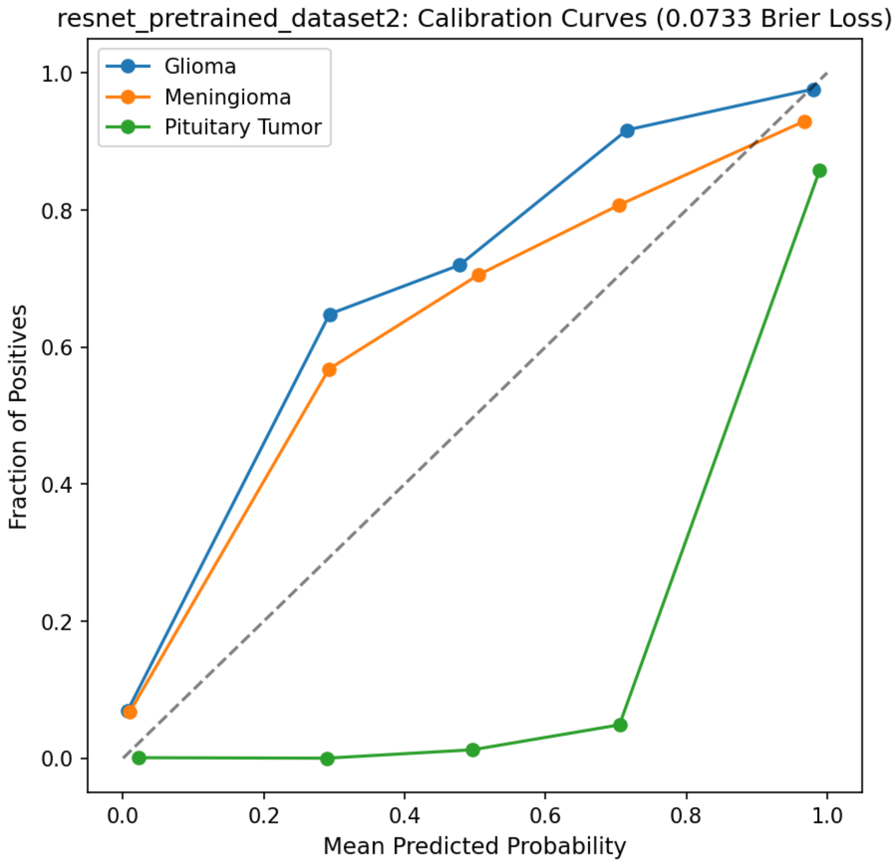
0.977
ROC-AUC
all classes ≈ 1.00

0.073
Brier loss
lower is better





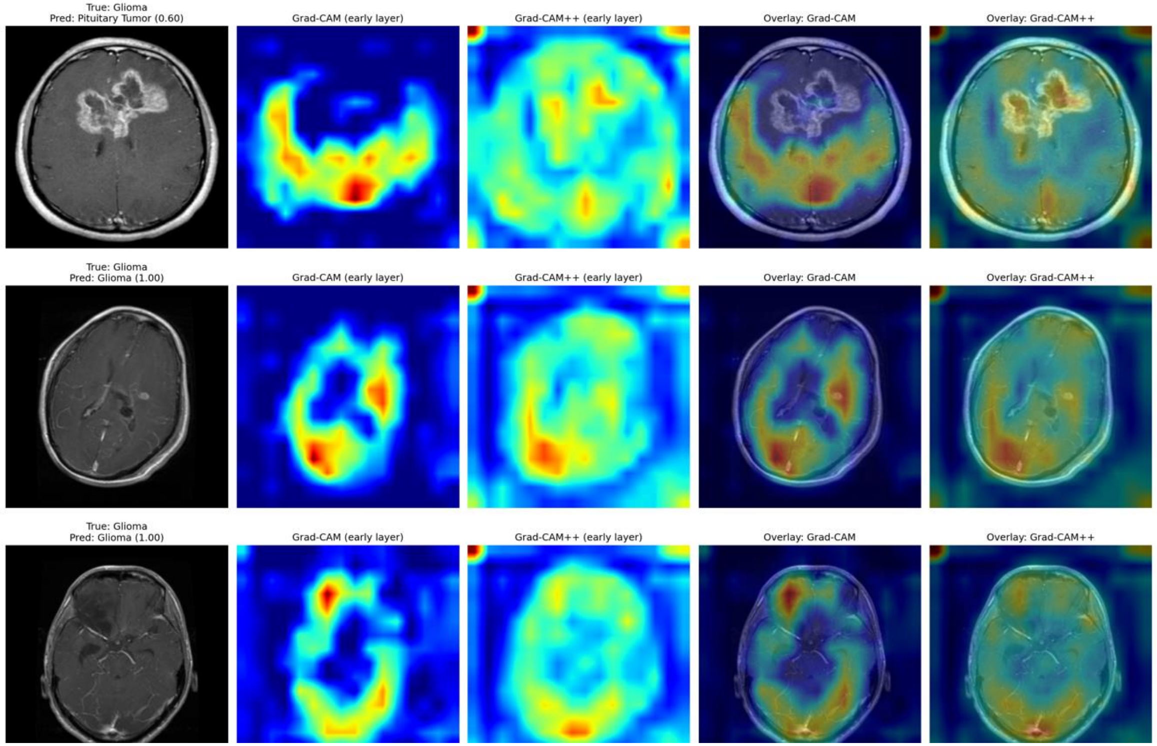
ROC and precision-recall curves



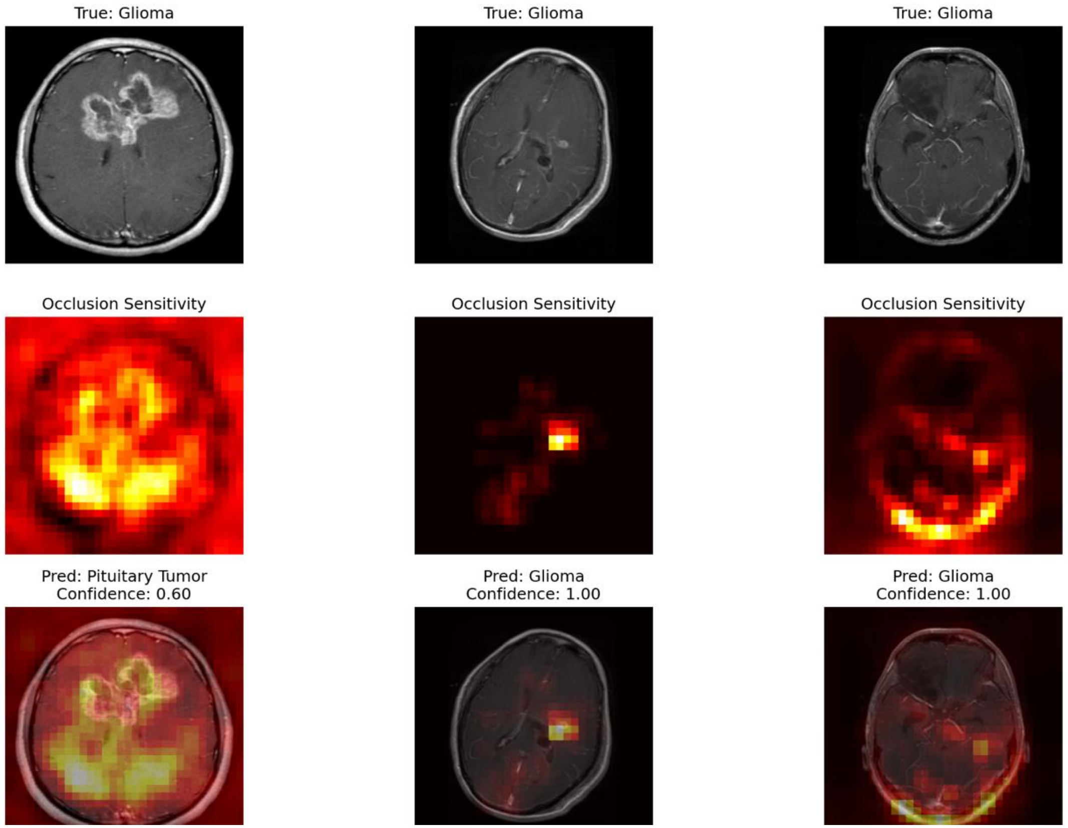
Calibration curve

Interpretability

Grad-CAM and occlusion were used to test whether the model looked at clinically reasonable regions.



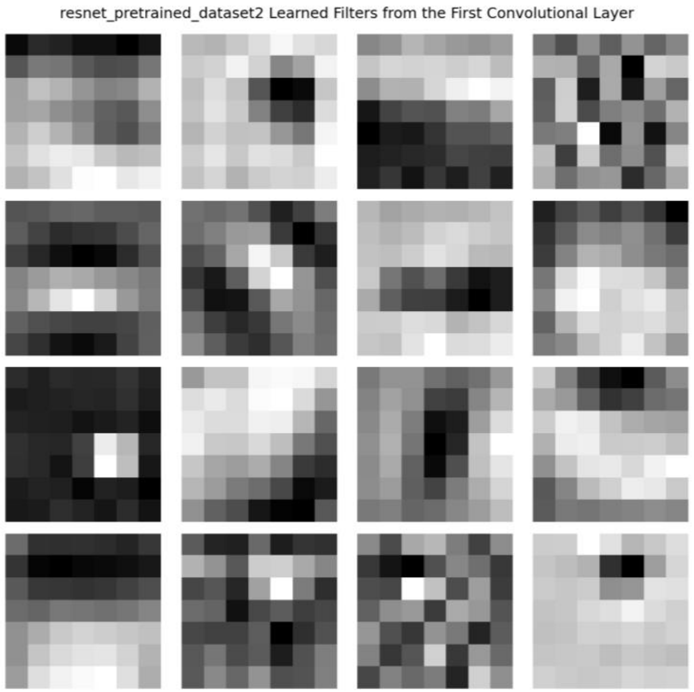
Grad-CAM on test dataset



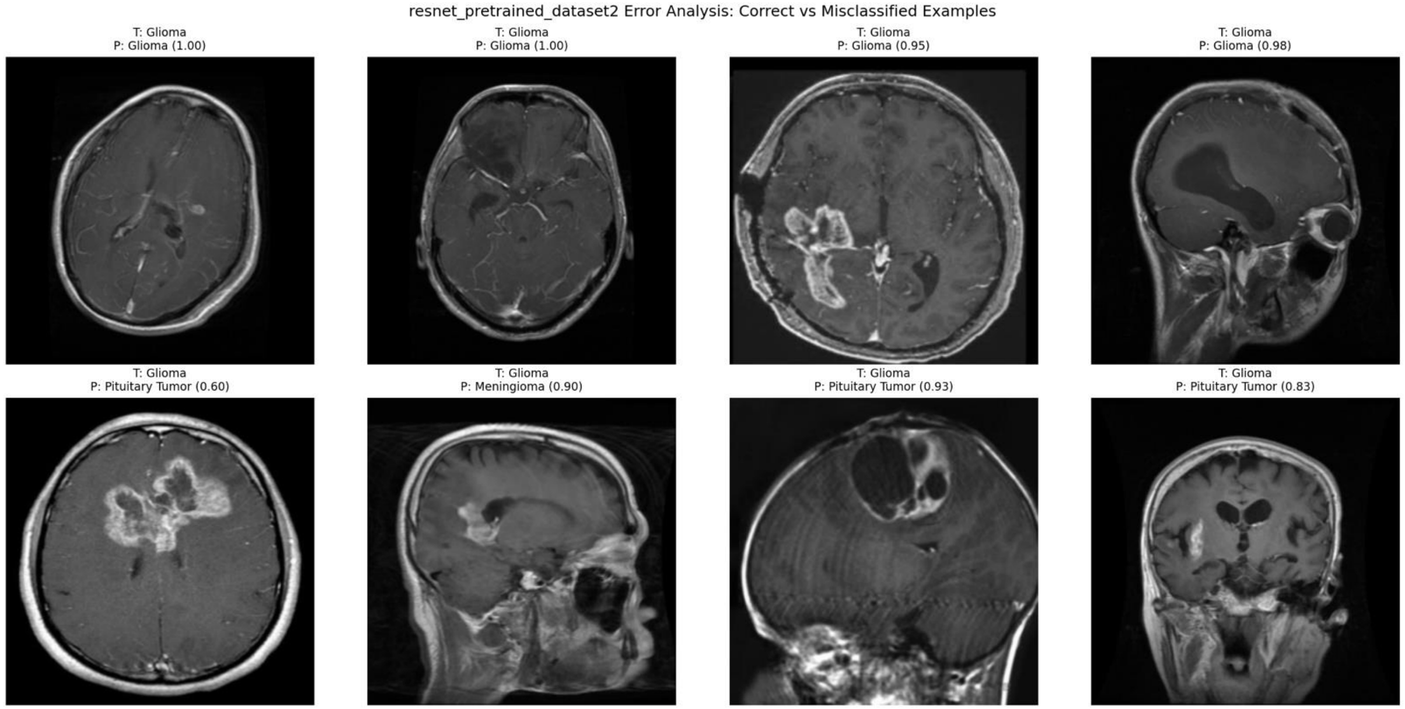
Occlusion on test dataset

Interpretability

Grad-CAM and occlusion were used to test whether the model looked at clinically reasonable regions.



Filters



Correct vs misclassified examples

The Governance

1

Acceptance testing

Local validation before use
Target: no unexpected performance drop

2

Routine monitoring

Track accuracy, confidence, class mix, and failure patterns
Cadence: weekly early, monthly after stable

3

Escalation

Pause if accuracy drops >5%, confidence shifts, or heatmaps behave unexpectedly

4

Human review

Clinician confirms, overrides, and documents final decision

Limitations define safe boundaries for use

The model should be used only within its validated scope.

Known limitations

- Single 2D slice input only.
- No clinical history or multi-sequence MRI information.
- Internal test performance only; external validation required.
- No segmentation masks available for quantitative heatmap validation.

Where performance may degrade

- Different scanner vendors or protocols.
- Low-quality, cropped, or rotated images.
- Uncommon tumor subtypes or postsurgical changes.
- Patient populations not represented in training data.

Should NOT be used for

- Standalone diagnosis.
- Replacing radiology review or pathology.
- Treatment planning decisions.
- Automated triage without human confirmation.

Conclusion

The most useful lessons came from calibration, error review, and interpretability checks.

Lesson learned

- Transfer learning is very effective for small-to-moderate medical imaging datasets.
- Calibration and error examples are needed because high accuracy can hide overconfidence.
- Grad-CAM is helpful but should not be treated as ground truth.
- Governance is part of the model, not an afterthought.

What should do differently

- Use full 3D MRI volumes or multiple slices instead of a single slice.
- Add clinical metadata and multi-sequence MRI if available.
- Collect expert tumor masks for localization validation.
- Run external validation on data from a different institution before any clinical pilot.

Thank you

Questions and feedback are welcome.